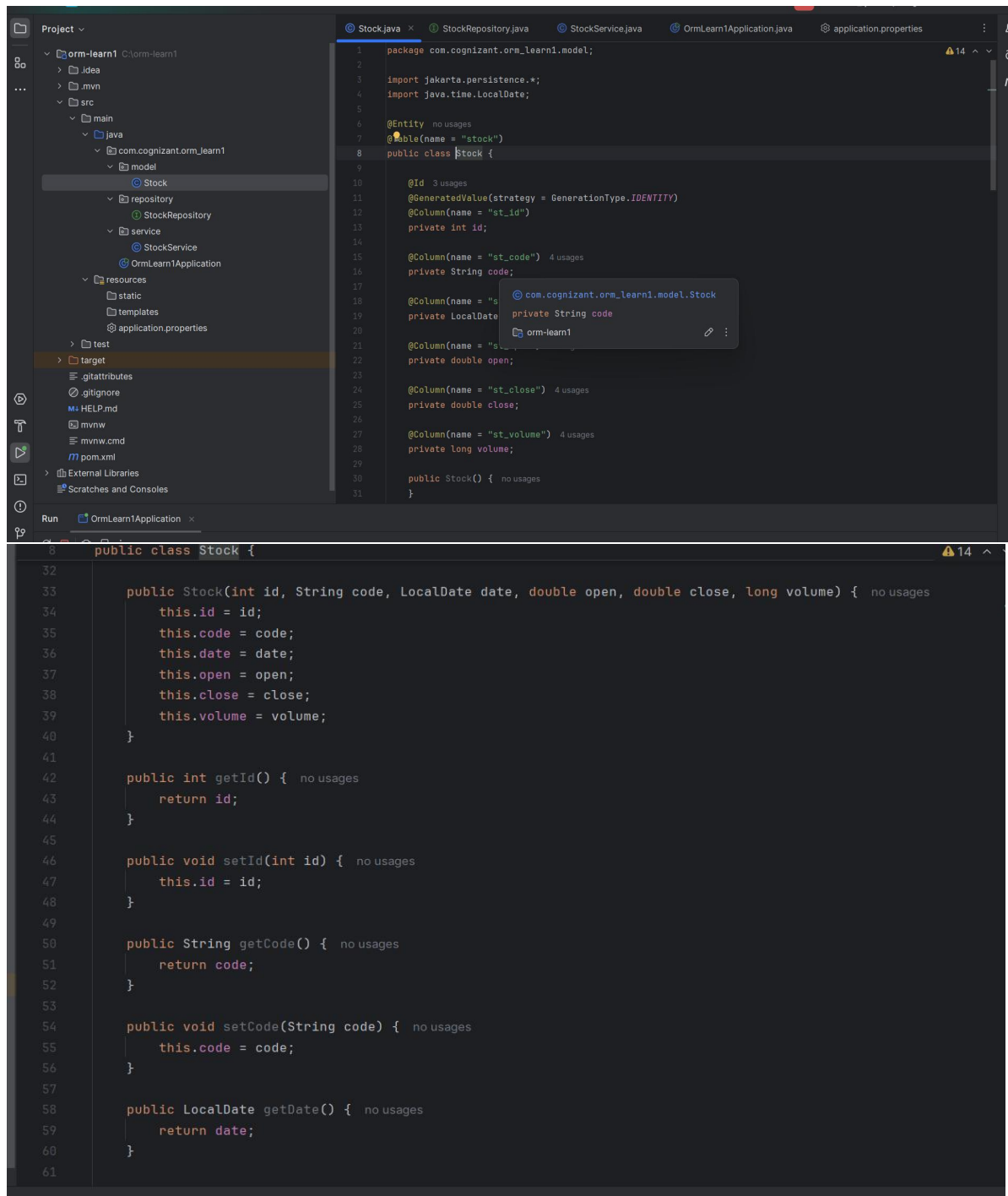


Spring Data JPA with Spring Boot, Hibernate

2.spring-data-jpa-handson



```
1 package com.cognizant.orm_learn1.model;
2
3 import jakarta.persistence.*;
4 import java.time.LocalDate;
5
6 @Entity
7 @Table(name = "stock")
8 public class Stock {
9
10     @Id
11     @GeneratedValue(strategy = GenerationType.IDENTITY)
12     @Column(name = "st_id")
13     private int id;
14
15     @Column(name = "st_code")
16     private String code;
17
18     @Column(name = "st_date")
19     private LocalDate date;
20
21     @Column(name = "st_open")
22     private double open;
23
24     @Column(name = "st_close")
25     private double close;
26
27     @Column(name = "st_volume")
28     private long volume;
29
30     public Stock() {
31
32
33     public Stock(int id, String code, LocalDate date, double open, double close, long volume) {
34         this.id = id;
35         this.code = code;
36         this.date = date;
37         this.open = open;
38         this.close = close;
39         this.volume = volume;
40     }
41
42     public int getId() {
43         return id;
44     }
45
46     public void setId(int id) {
47         this.id = id;
48     }
49
50     public String getCode() {
51         return code;
52     }
53
54     public void setCode(String code) {
55         this.code = code;
56     }
57
58     public LocalDate getDate() {
59         return date;
60     }
61 }
```

```

74     public double getClose() { no usages
75         return close;
76     }
77
78     public void setClose(double close) { no usages
79         this.close = close;
80     }
81
82     public long getVolume() { no usages
83         return volume;
84     }
85
86     public void setVolume(long volume) { no usages
87         this.volume = volume;
88     }
89
90     @Override
91     public String toString() {
92         return "Stock{" +
93             "code='" + code + '\'' +
94             ", date=" + date +
95             ", open=" + open +
96             ", close=" + close +
97             ", volume=" + volume +
98             '}';
99     }
100 }

```

Project view showing the project structure:

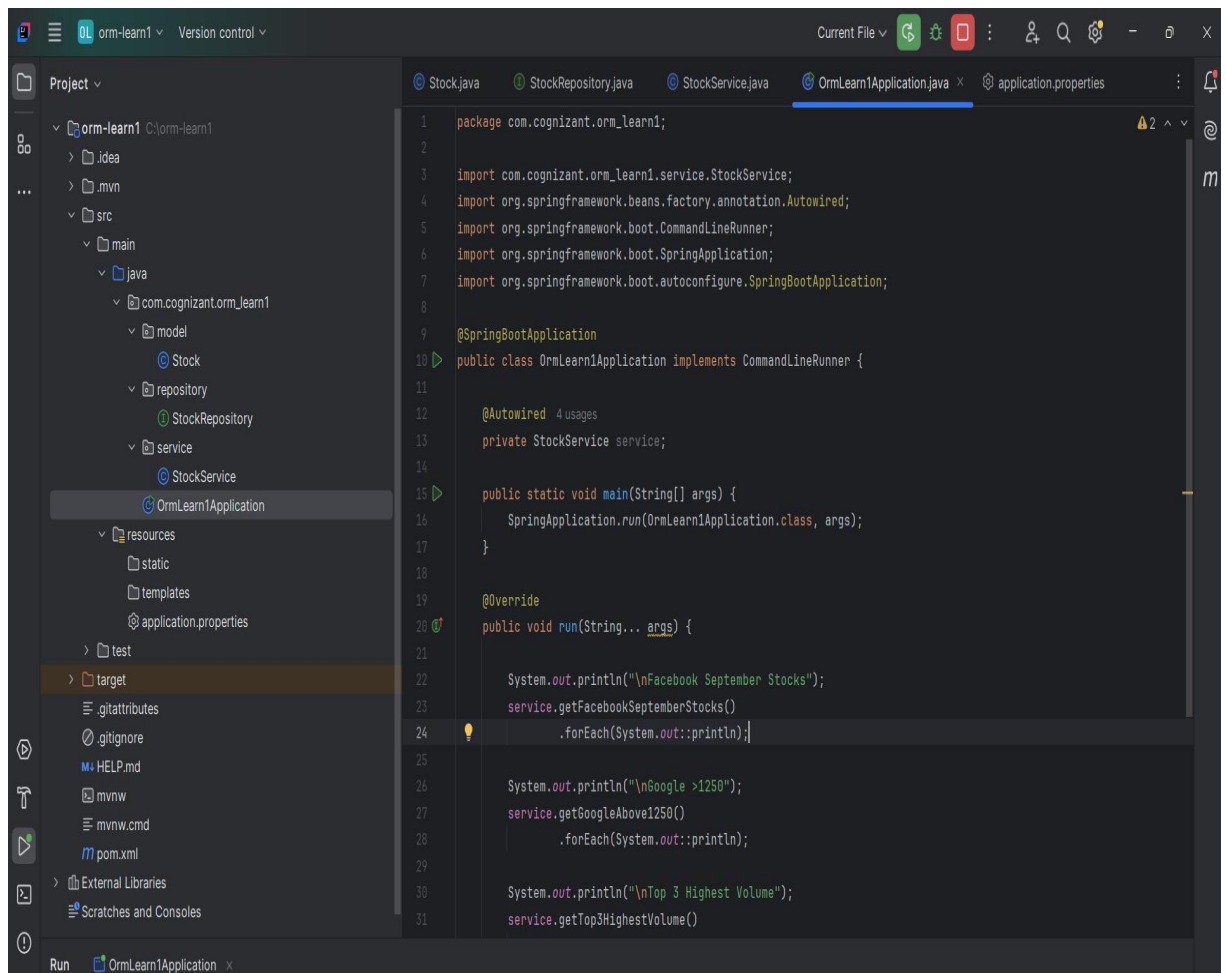
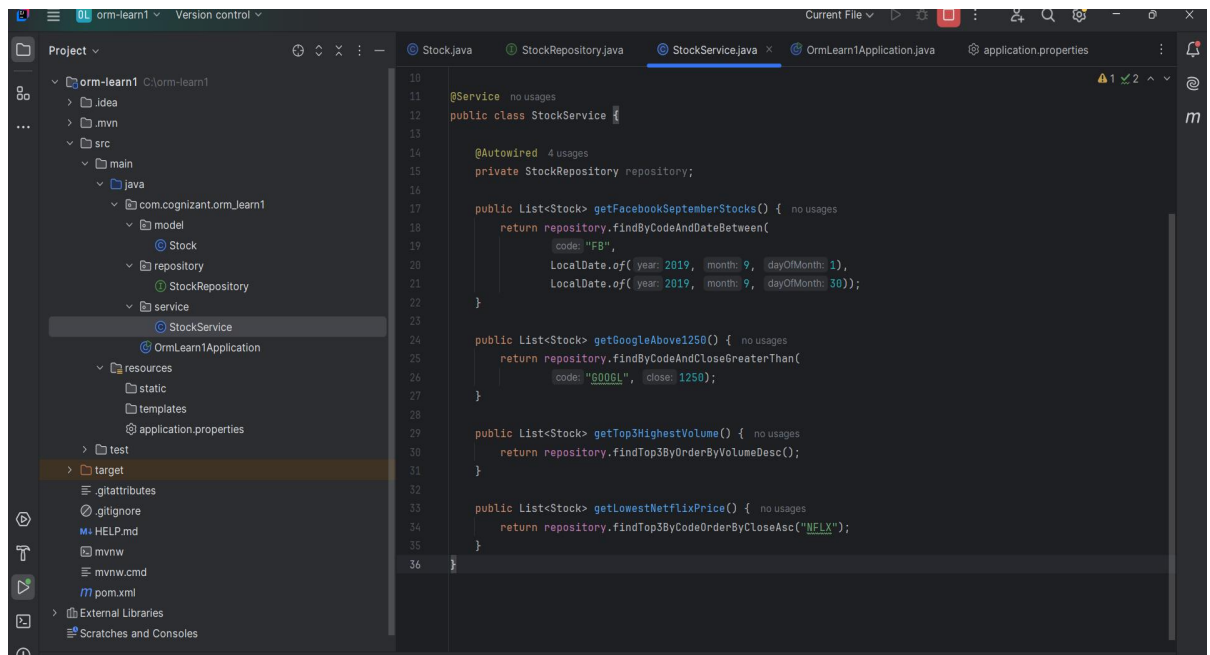
- orm-learn1
 - idea
 - mvn
 - src
 - main
 - java
 - com.cognizant.orm_learn1
 - model
 - Stock
 - repository
 - StockRepository
 - service
 - StockService
 - OrmLearn1Application
 - resources
 - static
 - templates
 - application.properties
 - test
 - target
 - .gitattributes
 - .gitignore
 - HELP.md
 - mvnw
 - mvnw.cmd
 - pom.xml
 - External Libraries
 - Scratches and Consoles

Code view showing the StockRepository.java file:

```

1 package com.cognizant.orm_learn1.repository;
2
3 import com.cognizant.orm_learn1.model.Stock;
4 import org.springframework.data.jpa.repository.JpaRepository;
5
6 import java.time.LocalDate;
7 import java.util.List;
8
9 public interface StockRepository extends JpaRepository<Stock, Integer> { no usages
10
11     // Facebook stocks in September 2019
12     List<Stock> findByCodeAndDateBetween( no usages
13         String code,
14         LocalDate start,
15         LocalDate end);
16
17     // Google stocks where close price >1250
18     List<Stock> findByCodeAndCloseGreaterThan( no usages
19         String code,
20         double close);
21
22     // Top 3 Highest Volume
23     List<Stock> findTop3ByOrderByVolumeDesc(); no usages
24
25     // Lowest 3 Netflix Prices
26     List<Stock> findTop3ByCodeOrderByCloseAsc( no usages
27         String code);
28 }

```



```
MySQL 8.0 Command Line Cli x + v
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_ormllearn |
+-----+
| country              |
| stock                |
+-----+
2 rows in set (0.01 sec)

mysql> DESCRIBE stock;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra           |
+-----+-----+-----+-----+-----+-----+
| st_id      | int           | NO   | PRI | NULL    | auto_increment |
| st_code    | varchar(255)  | YES  |     | NULL    |                 |
| st_date    | date          | YES  |     | NULL    |                 |
| st_open    | double        | YES  |     | NULL    |                 |
| st_close   | double        | YES  |     | NULL    |                 |
| st_volume  | bigint        | YES  |     | NULL    |                 |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM stock;
+-----+-----+-----+-----+-----+-----+
| st_id | st_code | st_date   | st_open | st_close | st_volume |
+-----+-----+-----+-----+-----+-----+
| 1     | FB      | 2019-09-01 | 180     | 185      | 1000000   |
| 2     | FB      | 2019-09-15 | 186     | 188      | 1500000   |
| 3     | GOOGL   | 2019-09-20 | 1240    | 1265     | 1200000   |
| 4     | NFLX    | 2019-09-21 | 280     | 275      | 800000    |
| 5     | NFLX    | 2019-09-22 | 270     | 268      | 900000    |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```